SADb Explorer — Sensory Afferent Database — 943 papers (382 app-only, removed from Airtable) · 833 curated · 318 with PDF · 1910–2026

Snapshot of the Airtable "Sensory Feedback" corpus. Offline core; the Search tab adds live OpenAlex queries. Rebuild: `myo python SADb_audit/export_corpus.py && SADb_audit/build_citation_graph.py && SADb_audit/app/build_app.py`

Table | Pivot | Bubble map | Search (online)

Layout: Year × citations | Style: Neurons | View: All papers | Degrees: 2 | Left click action: Focus paper

Click = focus: network out to N degrees — **direct connections colored** (green triangles = papers citing the focus arrive excitatory; red circles = papers the focus cites), 2nd degree grayed. Left panel = focus paper; right panel = connections (click to hop). Esc = back.

Focus paper

Neurobiological Bases of Rhythmic Motor Acts in Vertebrates

Grillner · 1985 · 1085 citations

10.1126/science.3975635

In-corpus graph

13 references · 57 citing papers · 555 shown (2 degrees)

Animals

Vertebrates | Mammals | Lamprey

Afferents

—

Pathways

—

Actions

References map | Cited-by map | Details | × Un-focus

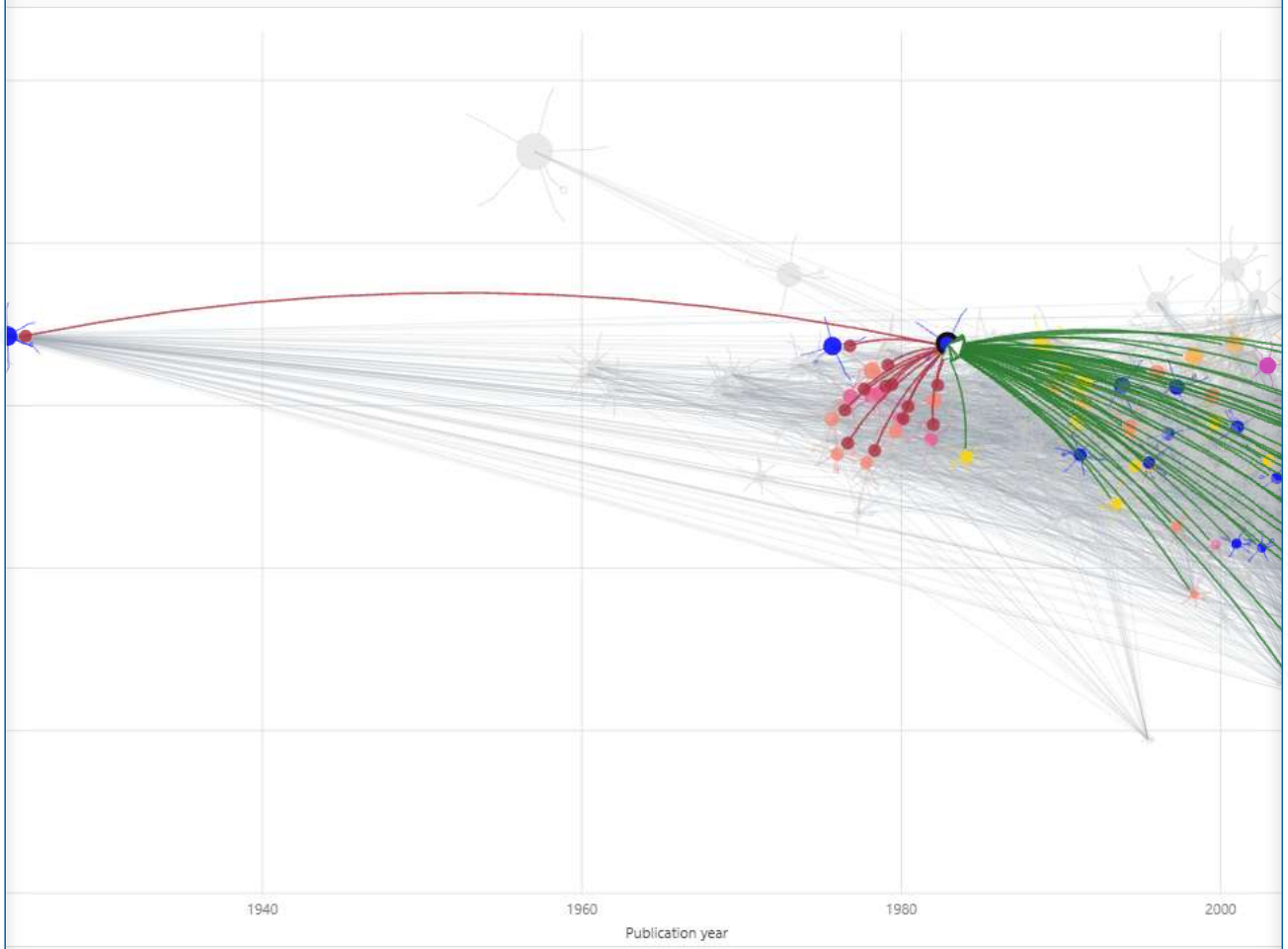
Curation note

Classic formulation of the principles underlying nervous control of innate rhythmic motor acts in vertebrates, establishing the in vitro lamprey spinal cord preparation in which the locomotor motor pattern can be elicited in isolated cord sections for detailed mechanism analysis, and proposing that parts of the locomotor network are reused in other motor acts, including learned ones.

Robot/sim translation

Implement modular CPG subnetworks shared across multiple motor programs in a simulated swimmer or walker and verify that reuse preserves timing and stability while reducing controller complexity.

obiological Bases of Rhythmic Motor Acts in Vertebrates · 555 papers within 2 degree(s)



Connections

▼ Cited by 57 (excitatory input)

Duysens et al. 2004
Sensory Influences on Interlimb Coordination During Gait

Rossignol 2011
Neural Control of Stereotypic Limb Movements

Fouad 2012
Comparative Locomotor Systems

Taga 1991
Self-organized control of bipedal locomotion by neural oscillators in un...

Taga 1995
A model of the neuro-musculo-skeletal system for human locomotion

Bässler 1986
On the definition of central pattern generator and its sensory control

Daun 2009
Control of oscillation periods and phase durations in half-center centra...

Ekeberg 1998
A neuro-mechanical model of legged locomotion: single leg control

Daun-Gruhn and Büschges 2011
From neuron to behavior: dynamic equation-based prediction of biological...

Grillner 1995
Neural networks that co-ordinate locomotion and body orientation in lamp...

Rossignol and Dubuc 1994
Spinal pattern generation

Tresch and Jarc 2009
The case for and against muscle synergies.

Minassian 2007
Human lumbar cord circuitries can be activated by extrinsic tonic input ...

Grillner 2006
Biological Pattern Generation: The Cellular and Computational Logic of N...